



Research Article

Stability of Wadsley-type vanadium oxides V_2O_5 and V_6O_{13} at high pressures

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ABSTRACT

A simple method to grow single crystals of α - V_2O_5 is presented and discussed. Crystal structures of two Wadsley phases, α - V_2O_5 ($Pmmn$, $Z = 2$) and α - V_6O_{13} ($C2/m$, $Z = 2$), are studied with single-crystal X-ray diffraction in diamond anvil cells up to 12.8 GPa and 9.4 GPa, respectively, at room temperature. α - V_2O_5 irreversibly amorphizes above about 7.3 GPa. No phase transition is observed at lower pressures. The polymorph that crystallizes from the amorphous material at high pressures and high temperatures is δ - V_2O_5 ($C2/c$, $Z = 4$) as evidenced by in-situ angle-dispersive synchrotron powder diffraction using large-volume multi-anvil press. No phase transitions were observed during high temperature decompression of the as synthesized δ - V_2O_5 . The crystal structure of α - V_6O_{13} is stable up to at least 9.4 GPa. Present data could be used to achieve better understanding of the high-pressure behavior of related Wadsley phases.

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1. Introduction

Binary vanadium oxides form a homologous series of Magnéli (V_nO_{2n-1}) and Wadsley (V_nO_{2n+1}) phases between V_2O_3 and VO_2 and between VO_2 and V_2O_5 , respectively [1,2]. The Wadsley phases undergo metal-insulator phase transitions (MIT) accompanied by a significant decrease of magnetic susceptibility due to the formation of spin-singlet V^{4+} - V^{5+} pairs in the insulating phases. V_2O_5 is widely used in catalytic and electrochemical applications [3–5]. Recently, V_6O_{13} has been extensively studied as a cathode material for rechargeable Li [6–11], Mg [12], or Zn [13–15] ion batteries. It is considered an ideal cathode material due to its high theoretical capacity and energy density.

α - V_2O_5 ($Pmmn$, $Z = 2$) is built of VO_5 square pyramids sharing edges and corners (Fig. 1). The V atom is displaced toward the apex of the square pyramid due to the vanadyl $V=O$ double bond. The sheets are held together by weak interlayer vanadium-oxygen interactions [16]. A similar square pyramid coordination can be found in some bronzes that adopt essentially the same structure as α - V_2O_5

[17]. In other bronzes, there is both 5- and 6-fold coordination, or only 6-fold coordination [18,19]. Deintercalation of $M_xV_2O_5$ bronzes ($M = Li^+$, Ag^+ , Mg^{2+} , Zn^{2+} , etc.) results in other V_2O_5 polymorphs [20–24].

α - V_6O_{13} undergoes a first-order metal-insulator phase transition at $T_c \approx 150$ K and exhibits an antiferromagnetic order at $T_N \approx 50$ K [25–28]. At room temperature, the structure ($C2/m$, $Z = 2$) is composed of edge- and corner-sharing VO_6 octahedra arranged in alternating single and double layers (Fig. 1). The symmetry of the low temperature phase in the temperature range $50 < T < 150$ K is disputed to be $C2$ [29], $P2_1/a$ [30], or Pc [31]. The crystal structure below T_N is assumed to be like that of the insulating phase. However, it has never been unambiguously characterized using single-crystal X-ray diffraction. The structure of α - V_6O_{13} at room temperature can be derived from α - V_2O_5 by first removing all atoms from every third oxygen atom plane along $\langle 001 \rangle$ and then by introducing crystallographic shear of $\frac{1}{2}[10\bar{3}]$ [32]. The shear in β - V_6O_{13} ($Cmma$, $Z = 4$), which can be obtained by exposure of α - V_6O_{13} to an electron beam [32] and by hydrothermal synthesis [33], is $\frac{1}{2}[01\bar{1}]$. Alternatively, the structures of the V_nO_{2n+1} and metastable VO_2 ($VO_2(B)$, $C2/m$, $Z = 8$) (Fig. 2) oxides could be derived from the fcc structure of VO_x ($x \approx 1$) by introducing ordered vacancies in the oxygen close-packing array [1]. Non-stoichiometric $V_6O_{13.27}$, with an excess of oxygen in the

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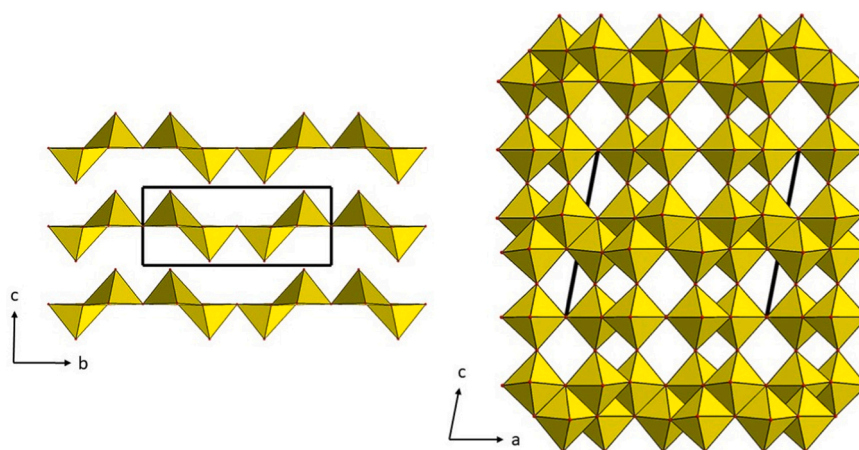


Fig. 1. Crystal structures of α - V_2O_5 (left) and α - V_6O_{13} (right) in different projections.

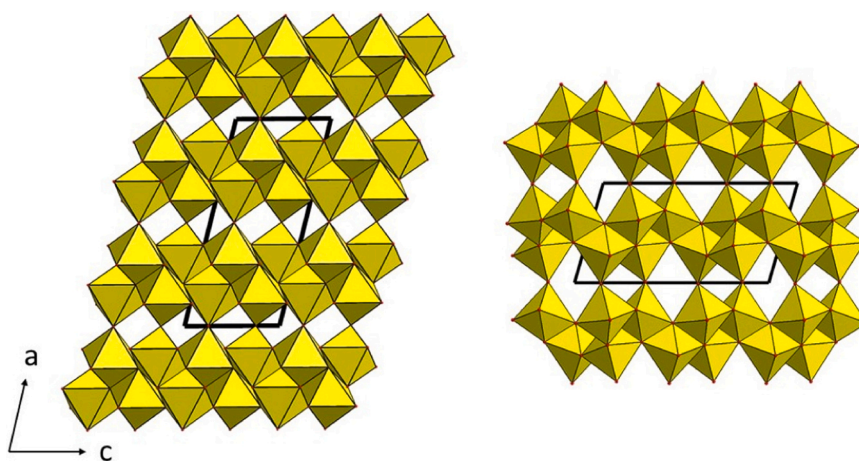


Fig. 2. Crystal structures of δ - V_2O_5 (left) and $\text{VO}_2(\text{B})$ (right) in the projection onto the (a,c) plane.

lattice, is an ionic conductor with anomalies in pressure-dependent resistivity at about 1–2 GPa and room temperature [34].

α - V_2O_5 is the only Wadsley phase whose crystal structure has been studied at high pressures. Investigation, using Raman spectroscopy at room temperature, has shown that a pressure-induced five-to-six coordination change of the V^{5+} cations occurs above 7 GPa and leads to a three-dimensional V_2O_5 lattice [35]. Upon decompression, intermediate phases have been recovered. A phase transition to an unidentified phase of V_2O_5 at about 6 GPa has also been detected in the synchrotron powder diffraction data [17]. In comparison, the analogous structural change in NaV_2O_5 , which has essentially the same structure as α - V_2O_5 , occurs at 25 GPa [17]. This indicates that intercalation results in a higher stability of the V_2O_5 layers under pressure. According to Arora et al. [36], α - V_2O_5 undergoes irreversible amorphization around 40 GPa. Upon decompression, the material exhibits several amorphous phases. Recently, a phase transition at 4.3 GPa has been detected using powder X-ray diffraction and Raman spectroscopy [37]. Upon further pressure increase, V_2O_5 starts to amorphize irreversibly at about 12 GPa and becomes fully amorphous at about 20 GPa. It must be pointed out that the quality of the data obtained so far in all the in-situ studies in diamond anvil cells has not allowed for any structure determination of the pressure-induced polymorphs of V_2O_5 . Moreover, the reported

pressures at which the material transforms or amorphizes are quite different. We believe that the reason is that up to now all the investigations have been carried out on powder samples of α - V_2O_5 at conditions beyond the hydrostaticity limits of the used pressure transmitting media [38,39]: CsI by Grzechnik [35], N_2 and a paraffin oil by Loa et al. [17], a 4:1 mixture of methanol and ethanol by Zou and Zhang [37]. Arora et al. [36] have performed their work without any medium at all.

Ex-situ high-pressure high-temperature synthesis has resulted in two new polymorphs of V_2O_5 . $\text{B-V}_2\text{O}_5$, isostructural with $\text{B-Nb}_2\text{O}_5$ (C2/c , $Z=4$) and formed at 8 GPa and temperatures above 1000 K [40], has been later renamed as δ - V_2O_5 [41,42]. The structure of δ - V_2O_5 has been reported by Balog et al. [41] and Zibrov et al. [42], from Rietveld refinements on the powder data of the recovered samples to ambient conditions. In δ - V_2O_5 , pairs of edge-sharing distorted VO_6 octahedra share corners with each other to form layers. Corner-sharing of the octahedra from adjacent layers results in a three-dimensional structure (Fig. 2). The second phase is β - V_2O_5 ($\text{P2}_1/\text{m}$, $Z=2$, $\beta \approx 90.1^\circ$), which has been prepared at $P=6.0$ GPa and $T=1073$ K [43]. That structure is built from distorted VO_6 octahedra that are linked by shared edges along [010] and by corner-sharing along [001]. This arrangement forms layers parallel to (100), held together by weak forces like in α - MoO_3 . β - V_2O_5 can be intercalated

with alkali metals [3,44]. The conditions at which the quenchable β and δ phases of V_2O_5 can be synthesized using large-volume presses have been compiled by Balog et al. [41].

Using in-situ energy-dispersive X-ray diffraction in a large-volume press, Kusaba et al. [45] have also identified two polymorphs of V_2O_5 at high pressures and temperature. The lattice parameters of the orthorhombic phase, observed at 4.9 GPa and about 600 K, correspond to those of β - V_2O_5 [41,43]. The second phase, formed at 10.1 GPa and about 700 K, has not been obtained pure and thus has not been identified.

In this study, we present a comparatively simple procedure to obtain single crystals of α - V_2O_5 . Since the discrepancies in the previous reports on the existence of pressure-induced phases in the α - V_2O_5 powder samples in the pressure range 4–6 GPa [17,37] and on the conditions at which V_2O_5 amorphizes [36,37] could be related to non-hydrostaticity in earlier experiments, a synchrotron single-crystal study on α - V_2O_5 using the pressure transmitting medium, which is hydrostatic to 10 GPa, is performed at room temperature. As we observed pressure-induced amorphization above about 7.3 GPa, an in-situ high-pressure high-temperature angle-dispersive synchrotron powder experiment in a large-volume multi-anvil press was also carried out to find out what crystallizes from the amorphous V_2O_5 and whether there are any phase transitions on pressure-temperature quenching to ambient conditions. In addition, we performed single-crystal diffraction measurements on related α - V_6O_{13} to examine its structural stability up to about 10 GPa at room temperature. The crystal structure of this compound has not been investigated at high pressures before. The results are discussed in relation to the high-pressure behavior of structurally related compounds.

2. Experimental

Single crystals of α - V_2O_5 were grown by fast cooling of a V_2O_5 melt. A V_2O_5 powder (Alfa Aesar 99.6%) was placed in an alumina crucible and then slowly moved during 15 min into the center of a tube furnace preheated in air to 973 K. The sample was kept at the hottest spot for 2 min and subsequently was moved out of the furnace in 15 min. After cooling, thin plate-like crystals were observed to have grown along their (*a*,*b*) plane [16]. Taking advantage of their exfoliation, mechanical stress was applied along the *a* and *b* axes of the formed plates. This simple treatment resulted in yellowish semi-transparent V_2O_5 single crystals with the lateral dimensions of about 1 mm × 5 mm and thickness of several micrometers. The crystals were then heated in air at 753 K for 1 h for relaxation of the internal stress [46]. Their quality was verified by single-crystal diffraction on a SuperNova diffractometer (Rigaku Oxford Diffraction Ltd) with a Mo-K α radiation ($\lambda_{\alpha1} = 0.70930$ Å, $\lambda_{\alpha2} = 0.71359$ Å, $U_{acc} = 50$ kV, $I = 0.8$ mA, $2\theta_{min} = 7.08^\circ$, $2\theta_{max} = 91.3^\circ$, $\Delta\phi = 1^\circ$, $t_{exp} = 1$ –7.5 s). The chemical composition was investigated by chemical analysis with Inductively Coupled Plasma - Optical Emission Spectroscopy (ICP-OES) using the spectrometer Thermo Scientific™ iCAP™ 7600. The obtained material was of a high purity (~99.6%) so that chemical techniques [47–51] for the removal of impurities were not performed.

High-pressure single-crystal diffraction measurements on α - V_2O_5 to 12.8 GPa were performed at the beamline P02.2 ($\lambda = 0.2905$ Å) of PETRA III (Hamburg, Germany) using a Perkin-Elmer detector (XRD1621). The crystal was loaded in a membrane driven BX90 diamond anvil cell [52]. Optical inspection of the sample at different pressures was carried out using a microscope Leica M165C equipped with a digital camera DFC295.

Single crystals of α - V_6O_{13} used in this study were prepared by Kawashima et al. [53]. Their quality was checked at room

temperature and ambient pressure measurement on SuperNova diffractometer (Rigaku Oxford Diffraction Ltd) with a Mo-K α radiation ($\lambda_{\alpha1} = 0.70930$ Å, $\lambda_{\alpha2} = 0.71359$ Å, $U_{acc} = 50$ kV, $I = 0.8$ mA, $2\theta_{min} = 6.96^\circ$, $2\theta_{max} = 91.48^\circ$, $\Delta\phi = 1^\circ$, $t_{exp} = 5$ s.).

High-pressure x-ray diffraction experiments on α - V_6O_{13} up to 9.4 GPa at room temperature were carried out in Ahsbahs-type diamond anvil cell [54] on a STOE IPDS-II diffractometer (STOE & Cie GmbH) with a Mo-K α source. The diffraction data were integrated with the X-Area software [55]. Additional single-crystal diffraction patterns using several diamond anvil cells [54,56] were collected up to 7.2 GPa at P24 at Petra III synchrotron (Hamburg, Germany) on a four-circle diffractometer (EH2) with a marCCD165 area detector, $\lambda = 0.44279$ Å.

During all measurements in diamond anvil cells, the pressure transmitting medium was a 4:1 mixture of methanol and ethanol and the pressure was determined using the ruby luminescence method [57]. All synchrotron diffraction data were integrated with the CrysAlisPRO software [58] and subsequently refined with the program Jana2006 [59].

In-situ angle-dispersive x-ray powder diffraction experiments at high pressures and high temperatures were performed on the α - V_2O_5 powder (Alfa Aesar 99.6%) using the Voggenreiter multi-anvil press in the beamline ID06-LVP ($\lambda = 0.23393$ Å) at the European Synchrotron Radiation Facility (Grenoble, France). Sample-detector distance and offset calibrations for the high-resolution 2D detector Pilatus3X-900 kW CdTe were carried out using SRM660a LaB₆ as a standard. The diffraction patterns collected continuously on compression, heating, and decompression were analysed using the Fit2d software [60]. The sample (0.7 mm in diameter and 2.0 mm in height) was surrounded by a capsule made of hexagonal BN (hBN, 2.4 mm in diameter and 2.0 mm in height). hBN discs (0.5 mm in thickness) were used to close the capsule that was placed inside a graphite heater (2.9 mm in diameter) together with two MgO plugs (each 2.4 mm in diameter and 2.7 mm in height). Cylindrical diamond epoxy windows in the MgO/Cr₂O₃ octahedron and boron epoxy gaskets were used to complete the sample assembly. The assembly was compressed to 11 GPa and heated to 1300 K by passing the current through graphite. Small parts of the recovered sample were mounted on glass pins to collect *ex-situ* Gandolfi patterns on a laboratory single-crystal diffractometer STOE IPDS-II ($\lambda = 0.71073$ Å). The patterns were integrated using the software X-Area [55]. Rietveld refinements were carried out with the program Jana2006 [59].

The experimental and refinement details, refined structural parameters, and interatomic distances at different pressures are in the [supplementary material](#). Crystallographic information files (cif) are deposited at the Inorganic Crystal Structure Database (FIZ Karlsruhe) with the numbers CSD 2151011–2151020 (α - V_2O_5), CSD 2151021 (δ - V_2O_5), and CSD 2151080–2151085 (α - V_6O_{13}).

3. Results and discussion

The crystal of α - V_2O_5 gradually changes its color in compression to become black and non-transparent at 12.8 GPa (Fig. 3). It remains dark when decompressed to ambient conditions. No damage to the crystal is visible under an optical microscope. The inspection of the reconstructed reciprocal space (Fig. 4) clearly reveals that up to 7.32 GPa all the reflections could be indexed with the orthorhombic lattice with no violation of the extinction rules for space group *Pmmn*. At higher pressures, on decompression and as recovered, at ambient conditions, the collected diffraction patterns exhibited very weak, broad, and smeared reflections, which cannot be indexed and integrated, indicating an irreversible pressure-induced amorphization.

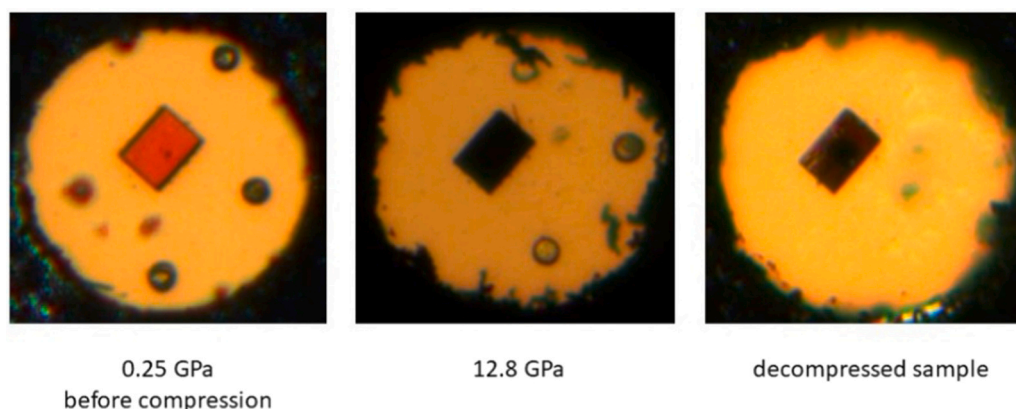


Fig. 3. The sample of V_2O_5 at different pressures during the single-crystal synchrotron measurements at Petra-III. The balls are rubies used to determine the pressure. The diameter of the hole in the gasket at 0.25 GPa before compression is about 0.15 mm.

The lattice parameters and unit-cell volume of α - V_2O_5 at atmospheric pressure are $a = 11.5125(4)$ Å, $b = 3.5614(2)$ Å, $c = 4.361(2)$ Å, and $V = 178.78(9)$ Å³. The change in the lattice parameters up to 7.32 GPa is highly anisotropic (Fig. 5). The compressibility is dominated by contraction of the c lattice parameter. The a lattice parameter slightly expands up to about 3.4 GPa and only contracts at higher pressures. All these observations on the axial and bulk compressibilities agree with those reported earlier [17,37]. The main effect of pressure on the crystal structure is the decrease of the interlayer V-O distance, resulting in a highly distorted octahedral coordination of the V atoms.

The P-V data could be fitted [61] with the third-order Birch-Murnaghan equation of state (Fig. 5): $V_0 = 178.7(5)$ Å³, $B_0 = 28(4)$ GPa, $B' = 23(5)$. Since the compressibility of α - V_2O_5 is highly anisotropic, B' was fixed at different values in the range 4–14 to yield bulk moduli B_0 between 53(10) GPa and 41(8) GPa in the previous studies on powder samples [17,37,45]. In the case of our single-crystal data, however, fixing B' at any value in this range results in a poor fit. In addition, the estimated standard deviations of the fitted $B_0 = 27(21)$ GPa and $B' = 24(32)$ with the zero-pressure volume constrained to the observed experimental value $V_0 = 178.78$ Å³ are too large so that the B_0 and B' parameters are not meaningful.

In earlier studies, the β - V_2O_5 and δ - V_2O_5 polymorphs have been identified from the quenched metastable samples synthesized at high pressures and high temperatures [40–43]. From the *ex-situ* experiments, it is not possible to deduce the actual synthesis and transformation paths of the material. The usual way to perform such experiments is to lower the temperature first and then to decompress the product at room temperature. In the case of V_2O_5 , the main questions are: (1) what crystallizes from the amorphous material,

(2) whether there are any phase transitions on lowering the temperature and pressure to ambient conditions. In our in-situ study using a large-volume multi-anvil press, V_2O_5 was compressed to 11 GPa and the amorphous material was recrystallized at high temperatures. The onset of crystallization was observed at about 700 K and 11 GPa (Fig. 6). The reflections due to the crystalline product can be indexed with a C centered monoclinic lattice. The product was recovered to ambient by following a path that crossed the conditions at which β - V_2O_5 and δ - V_2O_5 have been obtained in the earlier *ex-situ* studies [40–42], neither of which were observed. With no applied load, the final temperature (~ 500 K) was quenched. Rietveld refinement of the Gandolfi pattern of the recovered material demonstrated that the polymorph crystallizing from the amorphous phase is δ - V_2O_5 ($C2/c$, $Z = 4$) (Fig. 7).

The single-crystal measurements on stoichiometric α - V_6O_{13} provide evidence that it is structurally stable to 9.4 GPa, i.e., to the highest pressure reached in this study. There are no anomalies in our X-ray data that could be correlated with those observed at 1–2 GPa in the resistivity data on $V_6O_{13,27}$ [34]. The analysis of the reconstructed reciprocal space both from the laboratory and synchrotron data shows no hint towards a possible change of symmetry with increasing pressure (Fig. 8). It indicates that the structures observed at low temperatures at ambient pressure [29–31] do not occur on compression at room temperature. For the refinements of the synchrotron data with the program Jana2006, the lattice parameters, β angles, and unit-cell volumes were derived from those determined on the laboratory diffractometer using the respective axial and bulk compressibilities [61]. Additional refinements in non-centrosymmetric space groups $C2$ and Cm did not yield better-quality fits. Interatomic V-O distances, which include all three

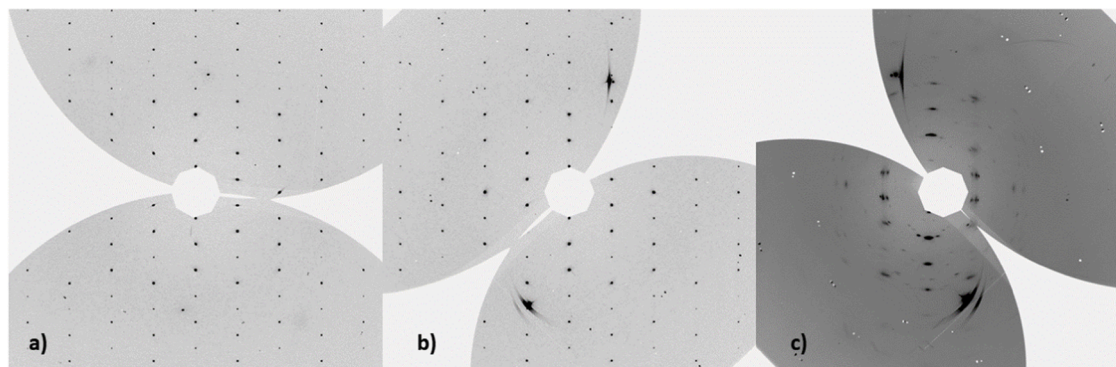


Fig. 4. Reconstructions of the reciprocal space of V_2O_5 in the $(h0l)$ projection at various pressures: a) ambient, b) 7.32 GPa, c) 9.57 GPa.

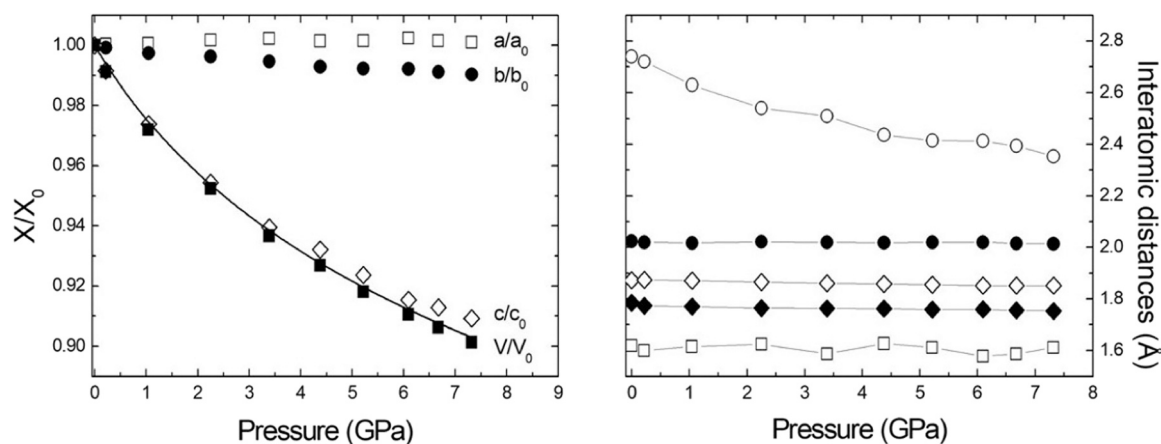


Fig. 5. Normalized lattice parameters and unit-cell volumes (left) and interatomic distances as a function of pressure (right) in V_2O_5 . The lattice parameters and unit-cell volumes are normalized to their values at ambient pressure (a_0 , b_0 , c_0 , V_0). The interatomic distances (from the top to bottom) correspond to V-O1 (interlayer), V-O2 (intralayer), V-O3 (interlayer), V-O1 (intralayer) bond lengths, respectively. The solid line in the left panel is the fit of the third-order Birch-Murnaghan equation of state. The lines in the right panel are guides for the eye. The estimated standard deviations are smaller than the size of the symbols.

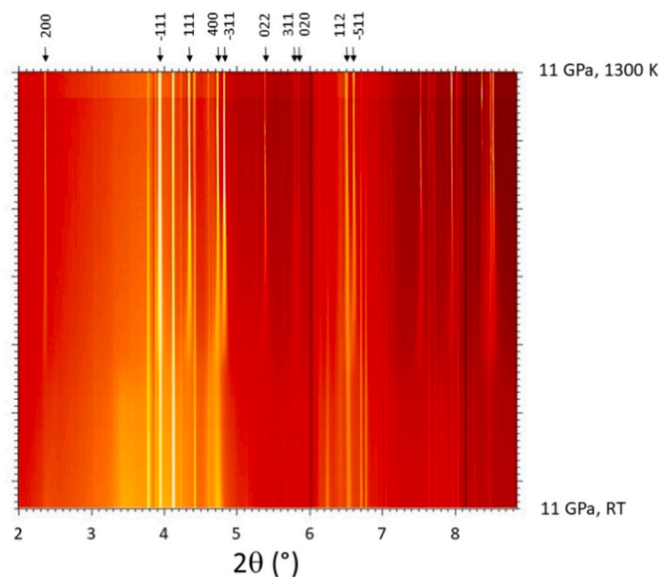


Fig. 6. Stacked powder patterns of V_2O_5 upon heating at 11 GPa. They show the progression of the normalized intensities of the patterns over the entire temperature range. Selected reflections of the δ polymorph are marked with the arrows and (hkl) Miller indices.

symmetry non-equivalent V atoms in the structure (V1, V2, V3), as a function of pressure are plotted in Fig. 9.

The lattice parameters and unit-cell volume of $\alpha-V_6O_{13}$ at atmospheric pressure are $a = 11.9150(5)$ Å, $b = 3.6756(2)$ Å, $c = 19.1390(5)$ Å, $\beta = 100.85(1)^\circ$, and $V = 436.10(4)$ Å³. The compressibility is only slightly anisotropic (Fig. 10) when compared to that of $\alpha-V_2O_5$ (Fig. 5). The P-V data for $\alpha-V_6O_{13}$ could be fitted with a third-order Birch-Murnaghan equation of state: $V_0 = 436.11(4)$ Å³, $B_0 = 131(4)$ GPa and $B' = 9(1)$. A comparison of the bulk moduli for $\alpha-V_2O_5$ and $\alpha-V_6O_{13}$ suggests that the structures of binary vanadium oxides with an octahedral coordination of vanadium atoms are less compressible than those with lower coordination numbers. It remains to be seen whether it is confirmed by the compressibilities of the other two Wadsley phases, V_3O_7 and V_4O_9 , in which the vanadium atoms have the coordination numbers 5/6 [62] and 4/5/6 [53],

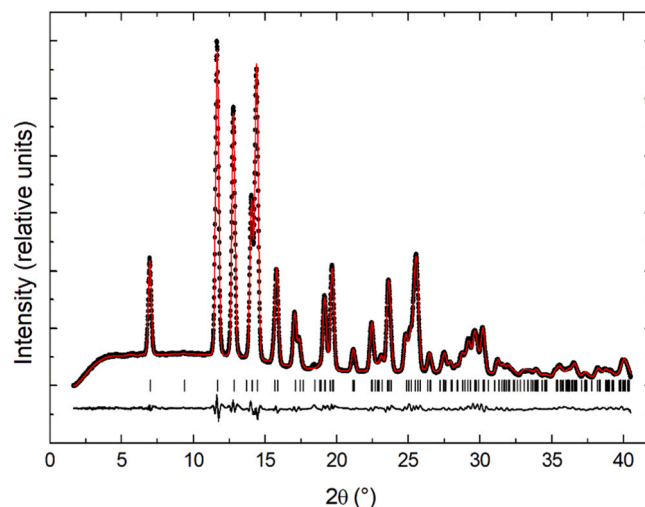


Fig. 7. Measured Gandolfi diffraction pattern (black circles) compared with the calculated (red lines) and difference (black lines) diagrams obtained using the Rietveld method for the recovered sample of $\delta-V_2O_5$ to ambient conditions: $C2/c$, $Z = 4$, $a = 11.944(3)$ Å, $b = 4.686(1)$ Å, $c = 5.314(2)$ Å, $\beta = 104.33(2)^\circ$, $R(F)_{\text{obs}} = 2.79\%$, $R_p = 2.82\%$, $wR_p = 3.84\%$. Vertical markers indicate the positions of the calculated Bragg reflections.

respectively. The bulk modulus for NaV_2O_5 is 24(3) GPa [17]. The bulk moduli of the Wadsley bronzes $\beta-Li_{0.33}V_2O_5$ ($B_0 = 62(3)$ GPa) [18], $\beta-Ag_{0.33}V_2O_5$ ($B_0 = 60(2)$ GPa) [18], and $\beta-Na_{0.33}V_2O_5$ ($B_0 = 64(6)$ GPa) [19], which all have distorted octahedra and square pyramids, are lower than that for $\alpha-V_6O_{13}$ but higher than for $\alpha-V_2O_5$.

Considering the crystal structures of $\alpha-V_6O_{13}$ and $\delta-V_2O_5$ shown in Figs. 1 and 2, one sees that the major difference between them is the absence of the single octahedral layer in the latter, similar to metastable $VO_2(B)$ [63] and derived from the VO_x arsitotype for the Wadsley phases [1]. The distorted octahedra have slightly different orientation with respect to each other but share their corners and edges within the double layers. When the single layer in $\alpha-V_6O_{13}$ is inspected, it turns out that it is nearly identical to the one in $\alpha-V_2O_5$ (Fig. 11), assuming that the vanadium atoms are six-fold coordinated to the oxygen atoms (despite one very long V-O distance). Consequently, the structure of $\alpha-V_6O_{13}$ could be described as being built of

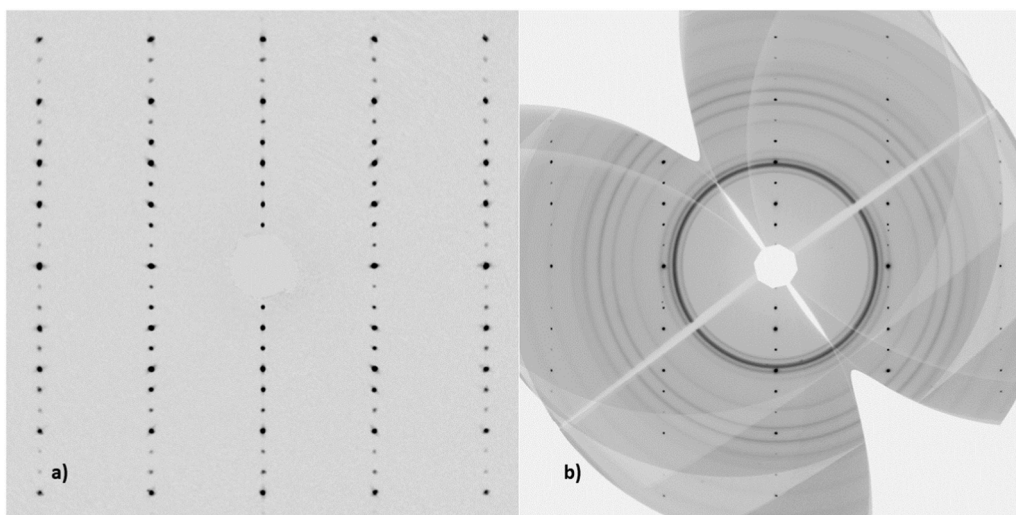


Fig. 8. Reconstructions of the reciprocal space of α - V_6O_{13} in the $(0kl)$ projection at various pressures: a) ambient, b) 7.2 GPa.

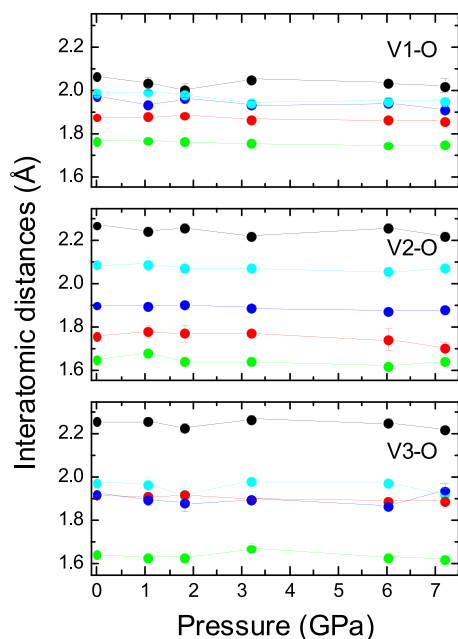


Fig. 9. Interatomic distances V-O in α - V_6O_{13} as a function of pressure. The estimated standard deviations are shown when larger than the size of the symbols. The lines are guides for the eye.

alternating $VO_2(B)$ - and V_2O_5 -like building blocks stacked along the c axis. The single layers in α - V_2O_5 , which share corners with each other, become more regular as pressure increases (Fig. 5). Such a structure is unstable upon compression and α - V_2O_5 amorphizes above 7.3 GPa. The δ - V_2O_5 polymorph, which crystallizes from the amorphous material (Fig. 6), consists of only double layers. From this point of view, δ - V_2O_5 is also very similar to $VO_2(B)$ [63] (Fig. 2). This clearly suggests that the binary oxides in the Wadsley series would tend to transform to the stable double-layer like configuration of the $VO_2(B)$ type at extreme conditions. However, this hypothesis needs to be confirmed by high temperature studies on α - V_6O_{13} and other V_nO_{2n+1} oxides at pressures higher than 9.5 GPa. The high-pressure behavior of metastable $VO_2(B)$ is also not known.

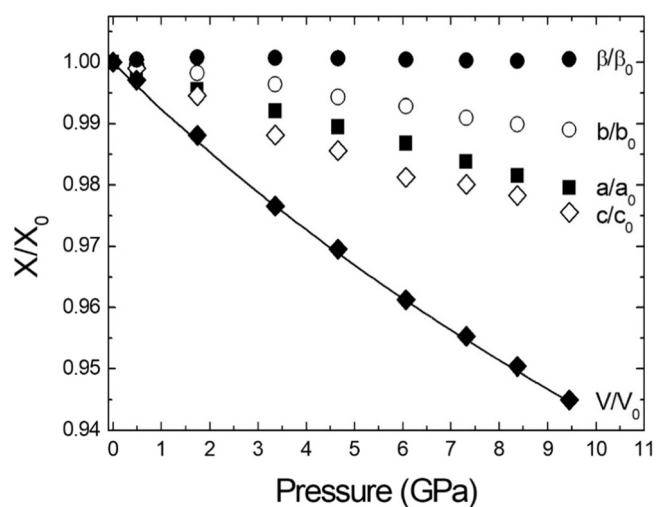


Fig. 10. Normalized lattice parameters and unit-cell volumes in α - V_6O_{13} as a function of pressure. The lattice parameters and unit-cell volumes are normalized to their values at ambient pressure (a_0 , b_0 , c_0 , β_0 , V_0). The solid line is the fit of the third-order Birch-Murnaghan equation of state.

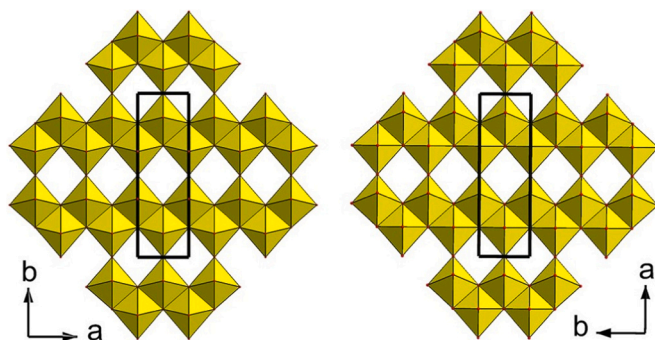


Fig. 11. Projections of the single octahedral layers in α - V_2O_5 (left) and α - V_6O_{13} (right) onto the (a,b) planes at ambient pressure. These layers have different orientation in the respective structures.

4. Conclusions

The new method to grow single crystals of α - V_2O_5 yields a high-quality material (Fig. 4) with a very fast, simple, and easily reproducible procedure. It is an improvement with respect to similar melting techniques [64–66] and overcomes the significant drawbacks (inhomogeneity, impurities, defects, long-time growth) of more complex chemical vapor transport (CVT) techniques [66,67].

In-situ studies on the high-pressure behavior and structural stability of the Wadsley phases α - V_2O_5 ($Pmmn$, $Z=2$) and stoichiometric α - V_6O_{13} ($C2/m$, $Z=2$) show the following:

- A single-crystal of α - V_2O_5 irreversibly transforms to an amorphous phase at about 7.3 GPa below the hydrostaticity limit of the 4:1 mixture of methanol and ethanol used as a pressure transmitting medium, i.e., at the pressure much lower than all those reported for powdered samples [36,37]. No other phase transitions are observed in the crystalline state.
- The polymorph that crystallizes from the amorphous V_2O_5 material is δ - V_2O_5 ($C2/c$, $Z=4$). There are no phase transitions observed on reducing pressure and temperature by a path designed to traverse those delineated in Balog [41].
- α - V_6O_{13} is structurally stable to at least 9.4 GPa. The structures observed at low temperatures and ambient pressure [29–31] do not occur on compression at room temperature.

CRedit authorship contribution statement

B. Viliam Hakala: Investigation, Formal analysis, Writing – original draft, Writing – review & editing. **Dimitra Manousou:** Investigation, Formal analysis, Writing – original draft, Writing – review & editing. **Konstantin Glazyrin:** Investigation, Formal analysis, Writing – review & editing. **Wilson A. Crichton:** Investigation, Formal analysis, Writing – review & editing. **Karen Friese:** Conceptualization, Investigation, Formal analysis, Writing – review & editing, Supervision. **Andrzej Grzechnik:** Conceptualization, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Supervision.

Authors contributions

All authors contributed to this work.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jallcom.2022.164966.

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